Project Report On

QuietSOS-Fall Detection and Alerting System

Submitted in partial fulfillment of the requirements for the award of the degree of

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By

Gokul Raj (U2203101)

Irene Ajith (U2203105)

Jefry Joseph (U2203111)

Jesel Gibi George (U2203112)

Under the guidance of

Ms. Ann Grace Attokaren

Department of Computer Science and Engineering
Rajagiri School of Engineering & Technology (Autonomous)
(Parent University: APJ Abdul Kalam Technological University)
Rajagiri Valley, Kakkanad, Kochi, 682039
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CERTIFICATE

This is to certify that the project report entitled ***"QuietSOS-Fall Detection and Alerting System"*** *is a bonafide record of the work done by* ***Gokul Raj (U2203101)***, ***Irene Ajith (U2203105)***, ***Jefry Joseph (U2203111)***, *and* ***Jesel Gibi George (U2203112)***, *submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science and Engineering" during the academic year 2022–2026.*

Project Guide
Ms. Ann Grace Attokaren
Assistant Professor
Dept. of CSE
RSET

Project Co-ordinator
Dr. Jincy J. Fernandez
Associate Professor
Dept. of CSE
RSET

Head Of Department
Dr. Preetha K G
Professor
Dept. of CSE
RSET

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Gokul Raj

Irene Ajith

Jefry Joseph

Jesel Gibi George

Abstract

QuietSOS is an AI-powered, vision-based emergency detection system designed to identify critical human conditions such as falls, fainting, and injuries that prevent individuals from recovering independently. The system emphasizes accurate, efficient, and context-aware detection while minimizing unnecessary computation and ensuring timely alerts. It operates using a two-stage intelligent framework. Initially, a lightweight YOLO-based prefilter continuously analyzes live video streams to detect human presence and monitor aspect ratio variations. Deeper processing is activated only when abnormal posture changes is observed, enabling selective inference that reduces computational load and supports real-time performance on low-power edge devices. Upon detecting a potential emergency, YOLOv8Pose extracts detailed skeletal keypoints representing body posture and movement. These pose sequences are then processed by a Long Short-Term Memory (LSTM) classifier to model temporal patterns, allowing the system to distinguish genuine emergencies—such as falls or fainting—from normal activities like sitting or bending. To improve reliability and reduce false positives, QuietSOS incorporates a short observation phase following suspected incidents. During this period, the system evaluates whether the individual remains down or unable to regain an upright position, indicating possible injury or unconsciousness. Once confirmed, real-time alerts are triggered using the Twilio API, sending SMS or call notifications along with precise location details to caregivers, security personnel, or emergency responders. The system also includes a centralized web-based monitoring dashboard that provides live camera feeds, structured event logging, and secure storage of incident video clips for verification and analysis. QuietSOS is suitable for environments such as offices, elevators, residential buildings, and assisted living facilities, where immediate human assistance may not always be available. Its applications include monitoring lone workers, detecting fainting incidents in confined spaces and enabling intelligent surveillance.

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Chapter 1

Introduction

Recent advances in computer vision and deep learning have enabled intelligent surveillance systems capable of analyzing human activities in real time. However, conventional surveillance systems still rely heavily on manual monitoring, which can delay the detection of critical incidents such as falls, fainting, or sudden collapses in locations where individuals may be alone. To address this issue, this project proposes **QuietSOS**, an AI-powered vision-based emergency detection system that automatically identifies fall events from surveillance video streams. The system uses a two-stage architecture consisting of a YOLO-based human detection and posture prefiltering stage followed by pose-based temporal analysis using YOLOv8-Pose and an LSTM network to reduce computational overhead. By detecting abnormal posture patterns and motion dynamics, the system can accurately recognize fall events and automatically send alerts to responders through SMS or call notifications.

1.1 Background

Surveillance cameras are widely deployed in public and private environments for monitoring and security. However, most traditional surveillance systems rely on manual observation, making it difficult to detect sudden medical emergencies such as falls or fainting, especially in environments where individuals may be alone. Falls are among the leading causes of accidental injuries worldwide, and delayed assistance can lead to severe health complications.

Many existing fall detection systems rely on wearable sensors such as accelerometers and gyroscopes. Although effective, these approaches require users to continuously wear the devices, which may be inconvenient or unreliable in real-world situations. Vision-based systems provide a non-intrusive alternative by analyzing human posture and movement directly from camera footage using deep learning models.

The proposed QuietSOS system adopts a two-stage vision-based framework to enable efficient and reliable fall detection. A lightweight YOLO-based detection stage continuously monitors human posture using simple geometric cues, while a second stage performs detailed pose-based analysis using YOLOv8-Pose and an LSTM network. This selective inference approach activates computationally intensive models only when abnormal posture changes are detected, thereby eliminating unnecessary computation while maintaining accurate real-time emergency detection.

1.2 Problem Definition

Many indoor environments such as offices, elevators and residential spaces lack automated systems to detect silent medical emergencies like falls, fainting, or sudden collapses. In such situations, individuals may be unable to call for help, leading to delayed medical assistance and increased risk of injury. The problem addressed in this project is the development of an intelligent vision-based system that continuously monitors video feeds to automatically detect fall-related incidents without relying on wearable devices or manual supervision, while accurately distinguishing genuine emergencies from normal human activities and generating timely alerts to responders.

1.3 Scope and Motivation

1.3.1 Scope

This project focuses on developing a real-time vision-based emergency detection system for indoor surveillance environments. The system analyzes video streams from fixed cameras to monitor human posture and motion patterns and detect fall or fainting incidents. By combining object detection, pose estimation, and temporal motion analysis, the system identifies abnormal posture changes and prolonged ground-level states that indicate potential emergencies and automatically generates alerts to enable faster response and improved safety monitoring.

1.3.2 Motivation

Medical emergencies such as falls or fainting often occur unexpectedly and may go unnoticed in unattended environments, resulting in delayed assistance and serious health consequences. The motivation behind this work is to develop an intelligent surveillance system that converts passive camera infrastructure into a proactive safety mechanism. By leveraging deep learning techniques for posture detection and temporal motion analysis, the system can autonomously recognize emergency situations and initiate alerts without user intervention, improving safety in workplaces, healthcare facilities, and residential environments.

Table 1.1: Comparison of Traditional vs. Proposed Systems

Aspect	**Traditional Systems**	**Proposed System**
Detection Type	Manual Monitoring	Automated AI-based Detection
Human Involvement	High	Minimal
Hardware Requirement	Wearable Devices / Sensors	Existing CCTV Cameras
Detection Approach	Motion Thresholds or Sensors	Pose and Temporal Analysis
Response Time	Delayed	Instantaneous
Alert Mechanism	Manual Reporting	Automatic Call/SMS Alert

1.4 Objectives

- To capture real-time video streams from surveillance or CCTV cameras.
- To detect human presence and monitor posture changes using a YOLO-based detection model.
- To apply lightweight visual prefiltering using bounding-box posture heuristics to identify abnormal posture changes.
- To extract skeletal keypoints using YOLOv8-Pose for detailed human posture representation.

- To construct temporal pose sequences and classify motion patterns using an LSTM neural network.
- To verify sustained ground-level posture to reduce false alarms.
- To automatically generate emergency alerts using communication services such as SMS and automated calls.

1.5 Challenges

Developing a reliable fall detection system presents several challenges. Variations in lighting conditions, camera angles, and environmental clutter can affect the accuracy of human detection and pose estimation models. Poor illumination or shadows may reduce the reliability of object detection algorithms, resulting in missed detections or false positives. Another challenge is distinguishing genuine fall events from normal activities such as sitting, bending, or lying down intentionally. Human posture changes during everyday activities can resemble fall-like movements, making accurate classification difficult. Additionally, rapid motion and partial occlusion can affect the extraction of skeletal keypoints, impacting temporal motion analysis. Another limitation arises when a person falls directly toward or away from the camera. In such cases, the bounding box aspect ratio may not change significantly because the width of the detected region does not increase, making abnormal posture detection more difficult. Furthermore, network latency or processing delays can affect the real-time pipeline, causing pose estimation to become slow or temporarily stuck, which may reduce the responsiveness of the system during continuous monitoring.

1.6 Assumptions

- The input video is obtained from a fixed-position surveillance camera.
- The monitored environment is an indoor space with moderate lighting conditions.
- Internet connectivity is available for sending alerts.
- The camera frame rate is sufficient to capture human motion clearly.

- The system is primarily designed for scenarios involving a single individual within the camera's field of view.
- Normal human activity is assumed to involve upright postures such as standing or walking.

1.7 Societal / Industrial Relevance

The proposed system can be deployed in environments where immediate detection of medical emergencies is critical. Examples include elderly care centers, office workplaces, elevators, and residential buildings. By enabling automated fall detection, the system can significantly reduce response time and improve safety outcomes for individuals at risk. Industries can integrate such AI-based monitoring systems into existing surveillance infrastructure to enhance workplace safety and employee well-being. In healthcare environments, the system can support caregivers by continuously monitoring patients and notifying staff when assistance is required. The ability to operate without wearable devices makes the solution practical and convenient for real-world deployment.

1.8 Organization of the Report

The remainder of this report is organized as follows:

- Chapter 1 introduces the background, problem definition, scope and motivation, objectives, challenges, assumptions and societal/industrial relevance of the proposed system.
- Chapter 2 presents a review of existing research and related work in vision-based fall detection systems.
- Chapter 3 describes the system architecture and methodology of the QuietSOS system, including detection models, pose analysis, temporal classification, and alert mechanisms.
- Chapter 4 presents the conclusion and summarizes the outcomes of the project along with potential future improvements.

1.9 Summary of Chapter

This chapter introduced the need for an intelligent system to detect silent medical emergencies such as falls and fainting in indoor environments. It discussed the limitations of traditional surveillance systems and wearable sensor-based approaches, highlighting the advantages of vision-based monitoring. The chapter also defined the problem addressed by the QuietSOS system and described its scope and motivation in improving safety through automated emergency detection. In addition, the objectives, key challenges, assumptions, and societal relevance of the system were presented, providing the foundation for the detailed system design and methodology discussed in the following chapters.

Chapter 2

Literature Survey

2.1 Introduction

Recent advances in artificial intelligence, deep learning, and human pose estimation have enabled researchers to design systems capable of accurately detecting fall events by analyzing body posture, motion patterns, and skeletal keypoints from video data. Many studies have proposed different approaches such as posture analysis, skeleton-based feature extraction, temporal motion modeling, and multimodal data fusion to improve the accuracy and reliability of fall detection systems. This section reviews several important research papers related to vision-based fall detection, highlighting their methodologies, key techniques, and major contributions.

2.2 Survey Papers

2.2.1 Comprehensive Review of Vision-Based Fall Detection Systems (Kwolek and Kepski, 2014) [1]

This paper presents a comprehensive survey of vision-based fall detection systems developed in recent years. The study systematically reviews existing techniques and categorizes them based on their feature extraction methods, classification models, and performance evaluation strategies. The authors analyze several fall detection approaches that use global features, local features, and depth-based information extracted from video sequences. The study compares both classical machine learning methods and modern deep learning approaches used for fall detection. The review highlights the increasing use of neural network models for improving detection performance under challenging conditions such as illumination variation, occlusion, and environmental noise. It also discusses widely used datasets and evaluation metrics for benchmarking fall detection algorithms. The paper identifies the limited availability of real-world fall datasets as a major challenge and provides recommendations for future research directions in vision-based fall detection systems.

2.2.2 A Two-Stage Fall Recognition Algorithm Based on Human Posture Features (Chen, Wei and Ferryman, 2012) [2]

This paper proposes a two-stage fall recognition algorithm designed to accurately distinguish fall events from normal daily activities. Human skeletal keypoints are extracted using the OpenPose framework, and posture-related features such as body deflection angles and spine ratio are computed to represent the inclination and orientation of the human body. The proposed system operates in two stages. In the first stage, human activities are categorized into stable, fluctuating, and disordered motion states to eliminate highly stable activities that are unlikely to be falls. In the second stage, temporal posture features are analyzed using machine learning classifiers, particularly Support Vector Machines (SVM), to identify fall events. Experimental results show that the two-stage framework improves fall detection accuracy by reducing false positives and effectively distinguishing falls from activities of daily living.

2.2.3 Fall Detection Based on Key Points of Human Skeleton Using OpenPose (Cao et al., 2019) [3]

This study proposes a fall detection approach that utilizes human skeletal keypoints extracted using the OpenPose pose estimation framework. The detection method relies on three important parameters derived from the skeleton keypoints: the descent velocity of the hip joint center, the inclination angle of the human body relative to the ground, and the width-to-height ratio of the bounding box surrounding the human body. These geometric features help determine whether a sudden change in body posture corresponds to a fall event. In addition to detecting falls, the system also analyzes whether the person is able to stand up after falling. This additional step helps reduce false alarms caused by normal activities such as sitting or lying down. Experimental validation conducted on recorded video sequences demonstrates that skeleton-based geometric feature analysis can achieve high accuracy in detecting falls while maintaining computational efficiency.

2.2.4 Sudden Fall Detection of Human Body Using Transformer Model (Zhang, Li and Chen, 2022) [4]

This research introduces a transformer-based framework for detecting sudden falls by analyzing temporal pose information extracted from video sequences. Human body keypoints are obtained using the MediaPipe pose estimation model, and the velocities of key joints are calculated to capture rapid changes in movement patterns that are typical during fall events. The transformer model employs an attention mechanism that focuses on important motion patterns over time, enabling the system to distinguish falls from normal human activities more effectively. By emphasizing the dynamic motion characteristics of body joints rather than only static postures, the method significantly reduces false alarms. Experimental results demonstrate high detection accuracy, showing that transformer-based temporal modeling is highly effective for real-time fall detection applications.

2.2.5 Human Fall Detection Using Pose Estimation (Yousuf et al., 2020) [5]

This paper presents a vision-based fall detection approach that utilizes human pose estimation to detect fall events from video sequences. The method focuses on extracting human skeletal keypoints using a pose estimation model and analyzing the spatial configuration of body joints. By monitoring the position and orientation of these keypoints, the system identifies abnormal body postures and sudden changes in body alignment that typically occur during a fall. The proposed method analyzes several posture-based features such as body inclination angle, relative joint positions, and changes in the height of the body's center. These features are used to distinguish fall events from normal daily activities such as walking, sitting, or bending. Since the approach relies on skeletal keypoints rather than raw image features, it reduces the influence of background noise, lighting variations, and other environmental factors. Experimental results demonstrate that pose-estimation-based fall detection can effectively identify fall events with high accuracy while maintaining computational efficiency. The study highlights that skeletal motion analysis provides a reliable solution for real-time fall detection in surveillance and healthcare monitoring systems.

2.3 Summary and Gaps Identified

2.3.1 Summary of Survey Papers

Table 2.1: Summary of Survey Papers

Paper Title	Advantages	Limitations
Comprehensive Review of Vision-Based Fall Detection Systems [1]	Provides detailed overview of existing approaches and datasets	Does not propose a new detection method
Two-Stage Fall Recognition Algorithm Based on Human Posture Features [2]	Improves accuracy by eliminating stable activities using two-stage classification	Depends heavily on posture features and may struggle with complex activities
Fall Detection Using Human Skeleton Keypoints (OpenPose) [3]	Uses geometric skeleton features and considers post-fall recovery	Limited robustness in crowded or occluded environments
Sudden Fall Detection Using Transformer Model [4]	Captures temporal motion patterns effectively using attention mechanism	Requires higher computational resources for real-time deployment
Human Fall Detection Using Pose Estimation [5]	Uses skeletal keypoints, making detection less affected by background and lighting changes.	Accuracy decreases if keypoints are not detected properly due to occlusion or camera angle.

2.3.2 Gaps Identified

1. Vision-based systems may face challenges such as occlusion, lighting variations, and complex backgrounds.

2. Some deep learning approaches require high computational power, making real-time deployment difficult.

3. Many existing systems focus only on detecting falls but do not provide immediate alert or notification to caregivers or emergency contacts.

4. Reducing false alarms while maintaining high detection accuracy remains a major challenge in fall detection research.

2.4 Summary of Chapter

This chapter presented a detailed review of several research papers related to vision-based fall detection systems. Different approaches including posture-based analysis, skeleton keypoint extraction, transformer-based temporal modeling, and multimodal feature fusion were discussed. Each method contributes to improving fall detection accuracy in different ways. The chapter also summarized the advantages and limitations of existing approaches and identified several research gaps that motivate the development of more reliable and efficient fall detection systems in future work.

Chapter 3

System Design

This chapter describes the design and architecture of the QuietSOS system. The system is designed to detect fall incidents in real time using computer vision and deep learning techniques. QuietSOS processes video streams from indoor surveillance cameras and analyzes human posture and motion patterns to identify abnormal events such as falls or fainting.

The design of the system follows a modular architecture where different components perform specific tasks such as video acquisition, human detection, posture analysis, temporal motion classification, and alert generation. This chapter presents the overall architecture, component design, module division, tools and technologies used, project responsibilities, and the development timeline.

3.1 Overall Architecture

The QuietSOS system follows a hierarchical two–stage processing architecture designed to detect fall incidents efficiently while minimizing unnecessary computation. The system continuously receives video frames from a fixed indoor surveillance camera and processes them through a lightweight prefilter stage followed by a deeper pose-based analysis stage when required[6].

In the first stage, a YOLO-based human detection model analyzes incoming frames and identifies the bounding box of the detected person. The width-to-height ratio of this bounding box is used to estimate the orientation of the human body. If the computed aspect ratio exceeds a predefined threshold, the posture is considered abnormal and the second stage of analysis is activated.

The second stage performs detailed pose-based temporal analysis. The system extracts skeletal keypoints using a pose estimation model and constructs temporal motion sequences. These sequences are analyzed using a Long Short-Term Memory (LSTM) neural network to determine whether the observed motion corresponds to a fall event. If a fall is detected and the individual remains on the ground for a sustained duration, the system triggers an emergency alert.

This two-stage architecture allows the system to operate efficiently by activating computationally intensive models only when abnormal posture changes are detected. The overall system architecture is shown in Figure 3.1.

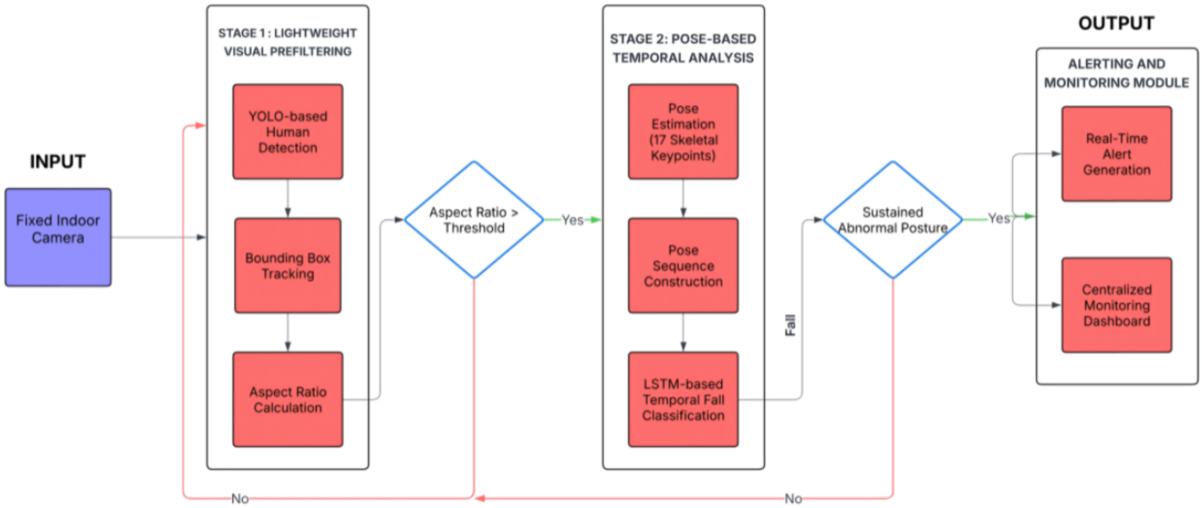

Figure 3.1: Overall System Architecture of QuietSOS

3.2 Component Design

The QuietSOS system consists of several interconnected components that work together to perform real-time fall detection. Each component performs a specific task within the processing pipeline.

3.2.1 Video Capture and Preprocessing

The video capture component is responsible for acquiring frames from the IP surveillance camera. A threaded video streaming mechanism is used to ensure that frames are continuously captured without blocking the main processing pipeline. The captured frames are then forwarded to the detection stage for further processing.

3.2.2 Human Detection Module

Human detection is performed using the YOLOv8 object detection model. The model analyzes each incoming frame and identifies human subjects present in the scene. The detector produces bounding boxes along with confidence scores indicating the probability of human presence. The bounding box coordinates are used to compute the width and height of the detected person, which are required for estimating posture orientation through geometric analysis.

3.2.3 Aspect Ratio Prefilter

The aspect ratio prefilter is responsible for detecting abnormal posture changes using bounding box geometry. The ratio between the width and height of the detected bounding box is computed to estimate body orientation.

A sigmoid function is applied to normalize the ratio and an exponential moving average is used to smooth sudden variations caused by detection noise. When the resulting aspect ratio score exceeds a predefined threshold, the posture is considered abnormal and the system activates the deeper pose analysis stage.

This prefilter stage helps eliminate unnecessary computation by preventing continuous execution of complex models when the person is standing or performing normal activities.

3.2.4 Lookback Frame Buffer

To ensure accurate temporal analysis, the system maintains a lookback frame buffer that stores previously captured frames. The buffer is implemented using a fixed-size queue capable of storing up to sixty frames.

During the normal monitoring phase, frames are continuously stored in this buffer. When the aspect ratio threshold indicates a potential abnormal posture, the system forwards the buffered frames to the pose estimation pipeline. This allows the system to analyze frames that occurred before the abnormal posture was detected. The lookback buffer ensures that the fall detection model receives the complete motion sequence leading to the fall rather than only the frames captured after the abnormal posture trigger.

3.2.5 Pose Estimation Module

The pose estimation module extracts skeletal keypoints using the YOLOv8-Pose model. The model detects seventeen keypoints representing major human joints such as the head, shoulders, elbows, hips, knees, and ankles. Each keypoint contains spatial coordinates and a confidence score indicating the reliability of the detection. These keypoints provide a detailed representation of human posture and body configuration.

3.2.6 Pose Sequence Buffer

The extracted keypoints from each frame are converted into feature vectors containing the coordinates and confidence scores of all detected joints[7]. These feature vectors are stored in a pose sequence buffer. The buffer maintains a sliding window of thirty frames that represents the temporal motion of the individual. Once the sequence buffer reaches the required length, the accumulated data is forwarded to the LSTM classifier.

3.2.7 LSTM Temporal Classification

Temporal motion analysis is performed using a Long Short-Term Memory (LSTM) neural network. The LSTM model processes the sequence of pose keypoints and learns temporal dependencies in human motion patterns[8]. The model outputs a probability score representing the likelihood that the observed motion corresponds to a fall event. If the probability exceeds the predefined fall detection threshold, the system marks the event as a potential fall candidate.

3.2.8 Ground Verification

To reduce false alarms, the system performs a post-event verification stage. The vertical position of the hip keypoints relative to the frame height is monitored to determine whether the individual remains on or near the ground. A fall is confirmed only if the person remains in a ground-level posture for a specified duration. If the person stands up or recovers before this duration elapses, the fall candidate is discarded and the system returns to normal monitoring mode.

3.2.9 Alert Generation

Once a fall event is confirmed, the system generates an emergency alert to notify responders[9]. The alert contains essential information such as the camera identifier and the timestamp of the detected incident. The notification is transmitted using the Twilio communication API, which enables automated alerts through SMS messages and phone calls.

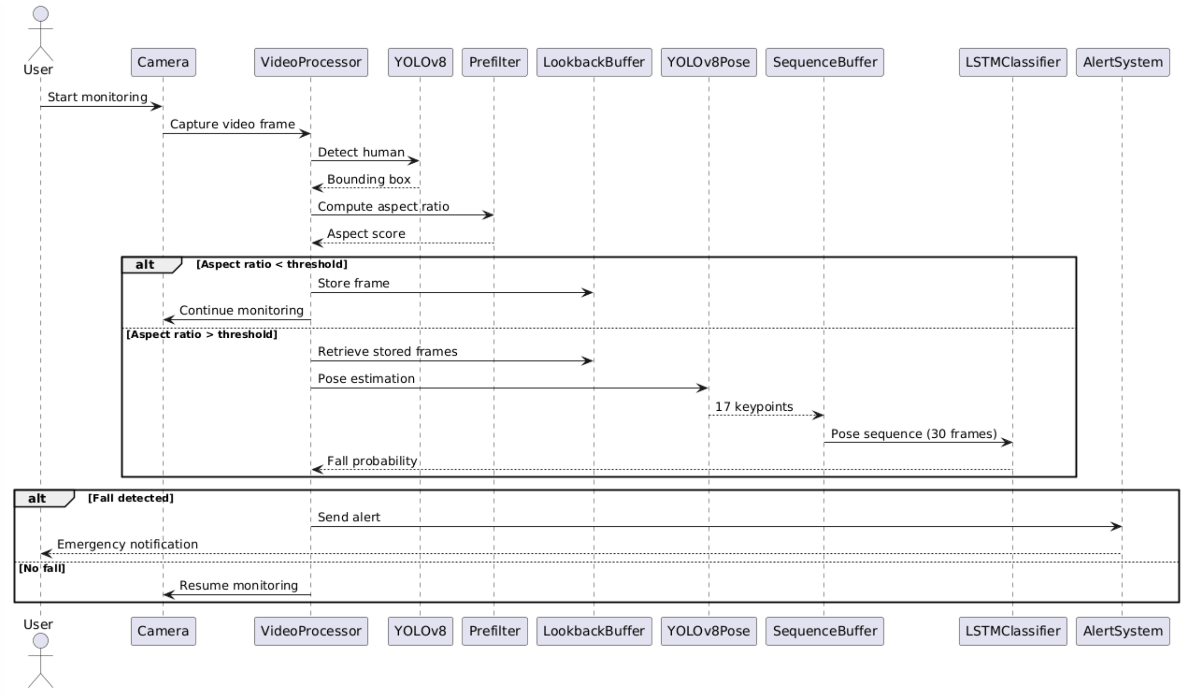

Figure 3.2: Sequence Diagram of QuietSOS System

3.3 Module Division

The QuietSOS system is divided into several functional modules that collectively perform real-time fall detection and emergency alert generation. Each module is responsible for a specific stage in the processing pipeline, enabling efficient analysis of video data and accurate identification of fall events.

- **Video Capture Module:** This module is responsible for acquiring live video frames from the IP surveillance camera. A threaded streaming mechanism is used to ensure continuous frame acquisition without interrupting the processing pipeline. The captured frames are forwarded to the detection stage for further analysis.

- **Human Detection Module:** The human detection module identifies human subjects present in the captured video frames. The YOLOv8 object detection model is used to detect persons and generate bounding boxes around them. These bounding boxes provide the spatial information required for posture analysis and further processing.

- **Posture Prefilter Module:** This module performs an initial posture analysis using bounding box geometry. The aspect ratio between the width and height of the detected person is computed to identify abnormal posture changes. When the aspect ratio indicates a potential abnormal posture, the system activates the deeper pose analysis stage.

- **Pose Estimation Module:** The pose estimation module extracts skeletal keypoints representing the major joints of the human body using the YOLOv8-Pose model. These keypoints provide a structured representation of human posture and are used for detailed motion analysis.

- **Fall Detection Module:** The fall detection module performs temporal analysis of pose keypoints using a Long Short-Term Memory (LSTM) neural network. By analyzing sequences of body movements across multiple frames, the model determines whether the observed motion corresponds to a fall event.

- **Alert Generation Module:** When a fall is confirmed, the alert generation module triggers an emergency notification[10]. The system sends alerts using the Twilio communication API, enabling automated notifications through SMS messages or phone calls to designated responders.

3.4 Tools and Technologies

3.4.1 Hardware

- NVIDIA GPU-enabled workstation for accelerating deep learning model training and inference

- IP surveillance camera for capturing real-time video streams

- Local processing workstation for running the fall detection system

3.4.2 Software

- Python programming language for implementing the fall detection system
- OpenCV for video capture and image processing
- Ultralytics YOLOv8 for real-time human detection
- YOLOv8-Pose for human pose estimation
- TensorFlow and Keras for developing the LSTM-based fall detection model
- NumPy for numerical computation and feature processing
- React.js for developing the monitoring dashboard
- Twilio API for sending emergency SMS or call alerts

3.5 Work Breakdown and Responsibilities

The project tasks were distributed among team members to ensure effective development and timely completion.

- **Gokul Raj** – Responsible for the development of the alert generation mechanism. Participated in performing and recording various fall scenarios that were used for testing.
- **Irene Ajith** – Involved in training the fall detection model and integrating the trained LSTM model and alerting mechanism with the overall system architecture. Also contributed to testing the system to ensure consistent and accurate fall detection and participated in documentation.
- **Jefry Joseph** – Involved in the model training process and worked on integrating the LSTM model into the complete system pipeline and enabling real-time fall detection. Also assisted in evaluating the system through testing under various fall scenarios and participate in documentation.
- **Jesel Gibi George** – Responsible for the development of the monitoring dashboard and connecting live video feed to dashboard. Also participated in acting out various fall scenarios that were used for testing and participated in documentation.

3.6 Key Deliverables

The main deliverables of the QuietSOS project include:

- Real-time fall detection pipeline
- Trained LSTM fall detection model
- Integrated pose estimation and detection system
- Automated alert generation mechanism
- Monitoring interface for incident tracking

3.7 Project Timeline

The project was carried out in multiple stages including research, system design, model development, implementation, and evaluation.

- **July 2025:** Initial project planning, literature survey on fall detection systems, and exploration of computer vision techniques for human activity recognition.
- **August 2025:** System architecture design and initial experimentation with human detection models. Development of the posture prefilter mechanism based on bounding box aspect ratio.
- **September 2025:** Implementation of pose estimation using YOLOv8-Pose and construction of pose feature representations for temporal analysis.
- **October 2025:** Development and training of the LSTM-based fall detection model to analyze pose sequences and identify fall events.
- **November 2025:** Implementation of the real-time video processing pipeline including video capture, human detection, and integration with the prefilter module.
- **December 2025:** Integration of pose estimation and LSTM fall detection into the real-time monitoring system.
- **January 2026:** Development of the automated alert generation mechanism and implementation of the monitoring dashboard for incident tracking.

- **February 2026:** System testing, performance evaluation, debugging, and preparation of the final project documentation.

3.8 Summary of Chapter

This chapter described the system design and architectural framework of the QuietSOS fall detection system. The overall architecture, component interactions, and module structure were presented. The chapter also explained the technologies used, the project responsibilities of each team member, and the development timeline. These design components provide the foundation for implementing the real-time fall detection system described in the following chapters.

Chapter 4

Model Training

This chapter describes the process followed to develop and train the fall detection model used in the QuietSOS system. The chapter first presents an overview of the dataset and the pose-based features used to represent human body movements. It then explains how the pose data is organized into temporal sequences suitable for machine learning models. The chapter further discusses the architecture of the Long Short-Term Memory (LSTM) network used for fall detection, along with the training strategy adopted for learning motion patterns associated with fall events. Finally, the evaluation methods used to assess the performance of the trained model are presented. The trained model forms an important component of the QuietSOS system and is integrated into the real-time monitoring pipeline for detecting fall incidents.

4.1 Dataset Preparation

The training dataset used in this project consists of pose keypoint data extracted from video frames. The pose estimation process identifies important human body joints such as the nose, eyes, shoulders, elbows, hips, knees, and ankles. Each frame contains the coordinates of these keypoints along with a confidence score that represents the reliability of the detected keypoint.

A total of 17 body keypoints are used in the model. For each keypoint, three values are recorded: the horizontal coordinate (X), vertical coordinate (Y), and the confidence score. These features collectively describe the posture of the person in a frame. The dataset is organized into two classes representing different activities:

- **Fall** – Sequences in which a person falls or collapses.
- **No_Fall** – Sequences in which the person is performing normal activities such as standing, walking, or sitting.

The pose information for each video is stored in CSV files where each row represents a detected keypoint for a particular frame. This structure allows the system to reconstruct the pose information for each frame and use it for temporal analysis.

Figure 4.1: A Single Frame From a Fall Video

Frame	Keypoint	X	Y	Confidence
1	Nose	887.0295	291.497	0.999612
1	Left Eye	893.7718	276.0556	0.999323
1	Right Eye	893.5267	274.7232	0.999274
1	Left Ear	893.2099	273.1479	0.999496
1	Right Ear	892.4576	275.3683	0.99953
1	Left Shoul	800.7289	296.3414	0.998381
1	Right Shou	801.7031	304.9897	0.999735
1	Left Elbow	812.8003	379.4524	0.092416
1	Right Elbo	822.4731	392.8804	0.981079
1	Left Wrist	859.8165	453.0038	0.156751
1	Right Wris	870.7602	470.7148	0.900513
1	Left Hip	669.9255	425.9225	0.998944
1	Right Hip	660.3382	433.7355	0.999057
1	Left Knee	618.1201	521.5059	0.398187
1	Right Knee	611.1808	533.0822	0.990843
1	Left Ankle	565.6684	631.8463	0.750626
1	Right Ankl	552.0197	651.05	0.993859
2	Nose	903.9511	300.6213	0.999574
2	Left Eye	909.2367	285.5034	0.999271
2	Right Eye	908.5958	284.2949	0.999214
2	Left Ear	907.7038	282.9214	0.999449
2	Right Ear	908.085	285.019	0.999487
2	Left Shoul	811.188	296.3412	0.998429
2	Right Shou	817.6697	308.8322	0.99973

Figure 4.2: CSV File Format

Since fall detection is a temporal event that occurs over a short period of time, individual frames are not sufficient to determine whether a fall has occurred. Therefore, sequences of frames are used as input to the model.

In this work, a sliding window approach is used to generate sequences from the pose data. Each sequence consists of 30 consecutive frames, which allows the model to observe the progression of body movement over time. A stride value of 5 is used to generate overlapping sequences, ensuring that important motion patterns are not missed.
For each frame, the pose features of all 17 keypoints are combined into a single feature vector. Since each keypoint contributes three values (X, Y, and confidence), the total number of features per frame is:

$$17 \times 3 = 51 \tag{4.1}$$

Thus, each input sequence provided to the model has the shape:

$$30 \times 51 \tag{4.2}$$

where 30 represents the number of frames in a sequence and 51 represents the pose features extracted from each frame.

4.2 Dataset Splitting and Class Balancing

After generating the sequences from the dataset, the data is divided into three subsets: training, validation, and testing. This separation ensures that the model can be trained effectively while also allowing unbiased evaluation. The dataset is divided as follows:

- **Training set** – 80% of the dataset used to train the model.
- **Validation set** – Used during training to monitor model performance and prevent overfitting.
- **Test set** – Used to evaluate the final performance of the trained model.

Since the number of fall and non-fall samples may not be equal, class weights are computed to handle class imbalance. These weights ensure that the model gives appropriate importance to both classes during training.

4.3 LSTM-Based Fall Detection Model

To capture temporal motion patterns, a Long Short-Term Memory (LSTM) neural network is used. LSTM networks are particularly suitable for analyzing sequential data because they can learn long-term dependencies between frames. The architecture of the proposed model consists of the following layers:

- **Masking Layer** – Handles padded values in the sequence data and ensures that missing keypoints do not affect model learning.
- **First LSTM Layer (64 units)** – Extracts temporal features from the input pose sequences.
- **Batch Normalization** – Stabilizes the training process and improves convergence.
- **Dropout Layer** – Reduces overfitting by randomly disabling neurons during training.
- **Second LSTM Layer (32 units)** – Further refines temporal feature extraction.
- **Dense Layer (32 neurons)** – Learns higher-level representations of the extracted features.
- **Output Layer** – A sigmoid activation function is used to classify sequences into fall or non-fall categories.

The final output of the network represents the probability that a given sequence corresponds to a fall event.

4.4 Training Strategy

The model is trained using the Adam optimization algorithm with a learning rate of 0.001. The binary cross-entropy loss function is used because the problem is formulated as a binary classification task. To improve the robustness of the training process, an early stopping mechanism is used. Early stopping monitors the validation loss during training and stops the training process if the performance does not improve for several epochs. This helps prevent overfitting and ensures that the best-performing model is retained. The model is trained for a maximum of 80 epochs with a batch size of 64. During training, the model learns temporal patterns associated with falling movements and distinguishes them from normal human activities.

4.5 Model Evaluation

After the training process is completed, the model is evaluated using the test dataset. Several performance metrics are used to measure the effectiveness of the fall detection model. These metrics help assess how accurately the system can identify fall events while minimizing false detections.

- **Accuracy** measures the overall correctness of the model's predictions. It represents the proportion of correctly classified samples out of the total number of samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{4.3}$$

  where TP represents True Positives, TN represents True Negatives, FP represents False Positives, and FN represents False Negatives.

- **Precision** indicates how many of the predicted fall events are actually correct. It measures the reliability of the fall predictions made by the system.

$$Precision = \frac{TP}{TP + FP} \tag{4.4}$$

  A high precision value indicates that the model produces fewer false alarms.

- **Recall** measures the ability of the model to correctly detect actual fall events. It represents the proportion of real fall events that are successfully detected by the system.

$$Recall = \frac{TP}{TP + FN} \tag{4.5}$$

  A high recall value indicates that the model is capable of detecting most of the fall incidents that occur.

These evaluation metrics provide a comprehensive understanding of the model's performance. Accuracy measures overall performance, precision evaluates the correctness of fall predictions, and recall measures the system's effectiveness in detecting true fall events.

4.6 Chapter Summary

This chapter described the development and training of the LSTM-based fall detection model for QuietSOS. It covered the use of pose keypoint sequences, the model architecture, and the training approach with class balancing and early stopping. Evaluation metrics including accuracy, precision, and recall were explained. The trained model forms a key component of the system, enabling accurate real-time fall detection. Overall, this chapter provides a clear understanding of how the model was designed, trained, and validated to reliably identify fall events.

Chapter 5

Results and Discussions

This chapter presents the experimental results and performance evaluation of the proposed AI-based fall detection system developed for the QuietSOS emergency monitoring platform. The system was trained using pose-based temporal sequences extracted from human activity data. Pose keypoints representing different human body joints were obtained for each frame and arranged into temporal sequences, allowing the model to capture motion patterns associated with fall and non-fall activities. The trained model was evaluated using commonly used classification metrics such as accuracy, precision, recall, and loss. These metrics help in assessing the ability of the model to correctly detect fall incidents while minimizing false alarms. The evaluation was conducted on a separate test dataset to ensure unbiased measurement of the model performance.

5.1 Evaluation of LSTM-Based Fall Detection Model

The proposed fall detection system utilizes a Long Short-Term Memory (LSTM) neural network to analyze sequences of pose keypoints extracted from video frames. LSTM networks are particularly suitable for this task because they can capture temporal dependencies and motion patterns across multiple frames. The dataset was divided into training, validation, and testing subsets to evaluate the generalization capability of the model. During training, the model learned to distinguish between two activity classes: Fall and No-Fall. The trained model was then evaluated on the test dataset using several performance metrics.

The evaluation results obtained from the trained model are summarized in Table 5.1.

Table 5.1: Performance of LSTM-Based Fall Detection Model

Metric	**Result**
Test Loss	0.32
Test Accuracy	88.47%
Test Precision	0.8612
Test Recall	0.8345

The model achieved an overall test accuracy of 88.47, indicating that the proposed LSTM network can effectively distinguish between fall and non-fall activities using pose-based temporal features. The relatively low test loss value of 0.32 suggests that the model has successfully learned meaningful representations from the input sequences and generalizes well to unseen data.

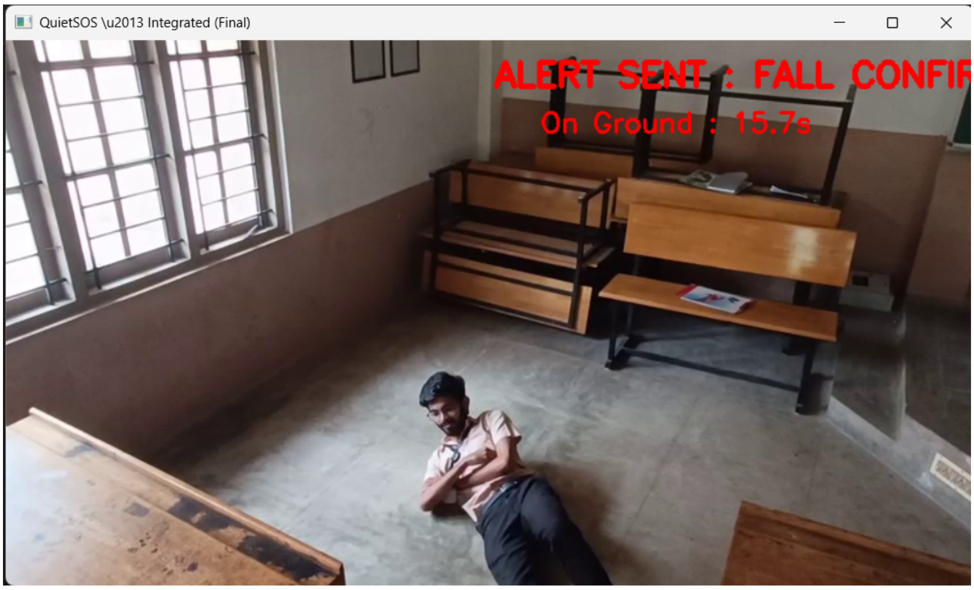

Figure 5.1: Alert sent when fall confirmed

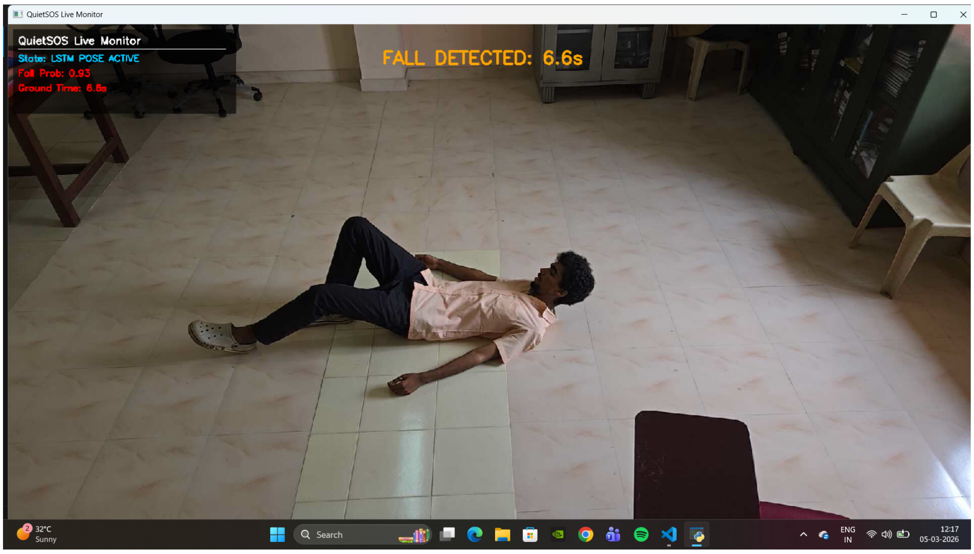

Figure 5.2: Post fall observation

5.2 Precision Analysis

Precision measures the proportion of correctly predicted fall events among all predicted fall events. The proposed model achieved a precision score of 0.8612, which indicates that approximately 86.12 of the predicted fall events are correct. This demonstrates that the system produces relatively few false alarms.

In practical emergency monitoring applications such as surveillance systems, maintaining a high precision is important to avoid unnecessary alerts and reduce alert fatigue among caregivers and monitoring personnel.

5.3 Recall Analysis

Recall measures the ability of the model to correctly detect actual fall incidents present in the dataset. The trained model achieved a recall value of 0.8345, which means that approximately 83.45 of real fall events were successfully detected by the system.

For emergency detection systems like QuietSOS, recall is a particularly important metric because missing a fall event could delay emergency assistance. The obtained recall value demonstrates that the proposed model is capable of identifying most fall incidents while maintaining a good balance with precision.

5.4 Effectiveness of Temporal Pose-Based Detection

The experimental results demonstrate the effectiveness of using pose-based temporal analysis for fall detection. Instead of relying solely on single-frame spatial features, the LSTM network analyzes sequences of body keypoints across multiple frames to understand motion dynamics. This approach enables the model to detect fall-related motion patterns such as:

- Sudden downward body movement
- Loss of balance or posture
- Rapid change in body orientation

By capturing these temporal characteristics, the system is able to distinguish fall events from normal daily activities more effectively.

5.5 Discussion

The obtained results indicate that the proposed system achieves reliable performance for fall detection tasks. The combination of pose keypoint extraction and LSTM-based temporal modeling enables the system to accurately analyze human movement patterns. Compared to traditional rule-based or threshold-based fall detection methods, the proposed approach offers several advantages:

- Improved ability to capture motion patterns across time
- Reduced false alarms due to temporal analysis
- Better generalization to different human activities
- Efficient integration with real-time surveillance systems

The achieved performance metrics demonstrate that the proposed system is suitable for integration into the QuietSOS emergency monitoring platform. When connected to real-time camera input, the system can continuously monitor human activity and automatically generate alerts when fall incidents are detected.

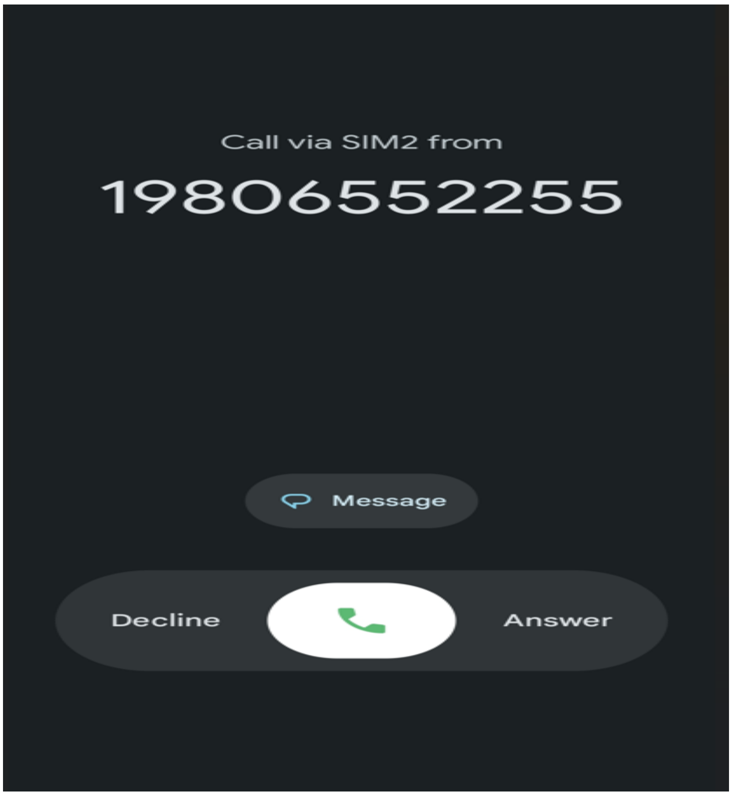

Figure 5.3: Call alert

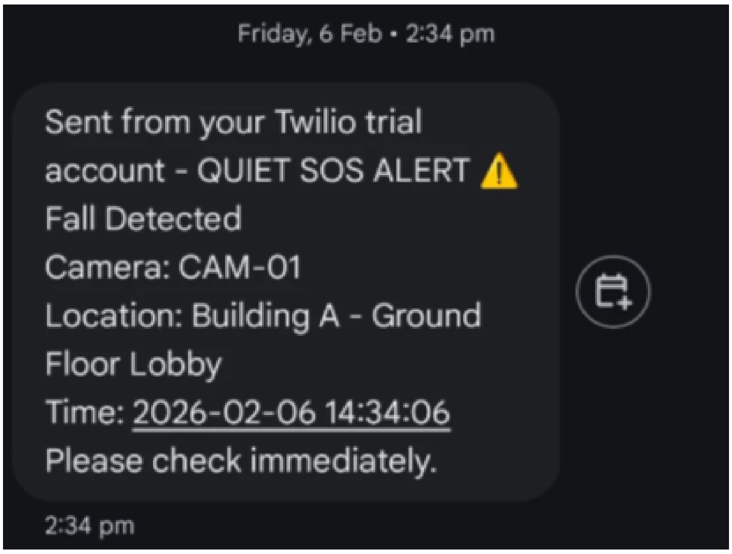

Figure 5.4: SMS alert

5.6 Chapter Summary

In summary, the proposed LSTM-based fall detection model achieved a test accuracy of 88.47, with a precision of 0.8612 and a recall of 0.8345. These results confirm that temporal analysis of pose keypoints is an effective approach for detecting fall events. The integration of the trained model into the QuietSOS system enables real-time monitoring and automated emergency alert generation, making the system suitable for applications such as elderly care and smart surveillance environments.

Chapter 6

Conclusions & Future Scope

6.1 Conclusion

The QuietSOS system presents an intelligent and efficient approach for real-time fall detection using computer vision and deep learning techniques. The system addresses the limitations of traditional vision-based fall detection methods that rely on continuous pose estimation, which can be computationally expensive and inefficient for real-time deployment. By introducing a two-stage architecture, QuietSOS first employs a lightweight YOLO-based human detection and posture prefilter to identify potentially abnormal body orientations. Only when suspicious posture patterns are detected does the system activate the more computationally intensive pose estimation and temporal analysis stage. This selective processing strategy significantly reduces unnecessary computation while enabling the system to operate effectively even in resource-constrained environments.In the second stage, the system utilizes YOLOv8 Pose to extract skeletal keypoints and constructs temporal pose sequences that are analyzed using a Long Short-Term Memory (LSTM) network to distinguish genuine fall events from normal daily activities such as sitting, standing, or walking. When a fall is confirmed, the system immediately triggers an alert mechanism that sends SMS notifications and voice call alerts through the Twilio API, while also logging the incident on a centralized monitoring dashboard for real-time supervision and event management. By combining lightweight visual prefiltering, pose-based temporal analysis, and automated alerting, QuietSOS provides a practical and scalable solution for monitoring indoor environments ultimately improving emergency response time and enhancing overall safety.

6.2 Future Scope

Future improvements to the QuietSOS system can focus on addressing the current limitations related to environmental conditions, scalability, and system robustness. One important direction is the integration of a multi-camera setup, which can provide multiple viewing angles to better capture body orientation and overcome the limitation where fall detection depends primarily on the bounding box width being greater than its height. Expanding the system to support multi-person environments is another significant enhancement, enabling reliable detection and tracking of multiple individuals simultaneously through advanced multi-object tracking and pose association techniques. Further improvements may include developing methods to handle occlusions, lighting variations, and camera placement issues through improved pose estimation models and adaptive preprocessing techniques. Additionally, optimizing the system for edge devices and improving network reliability for alert delivery can enhance scalability and ensure consistent real-time performance in diverse deployment environments.

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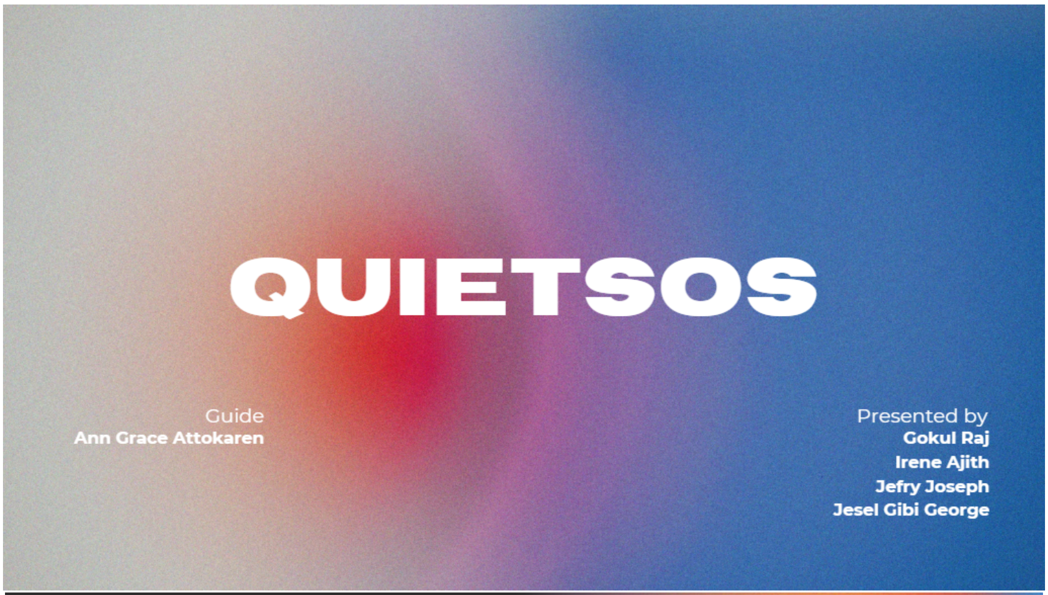
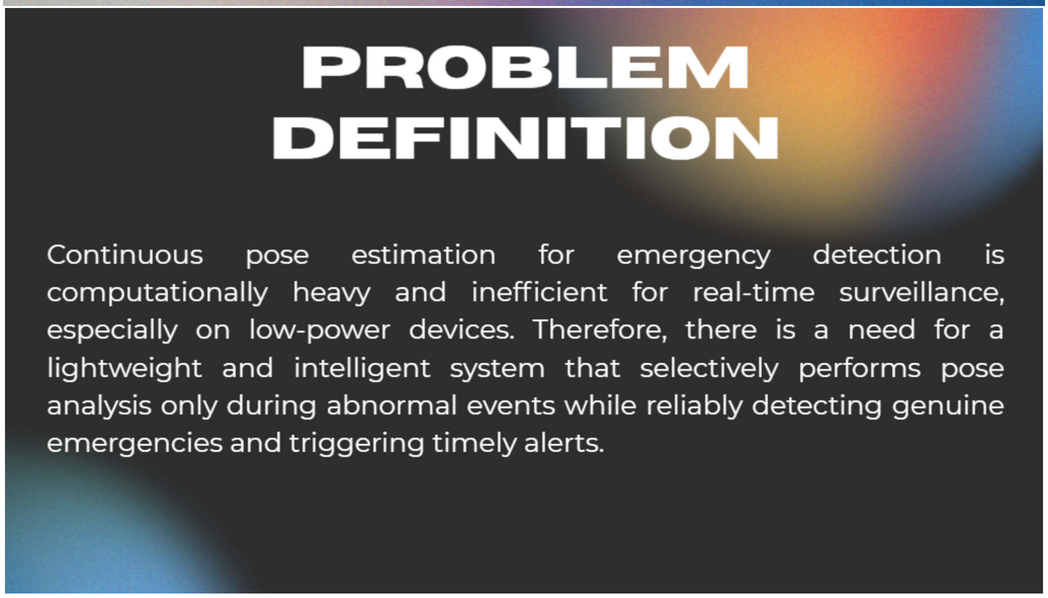
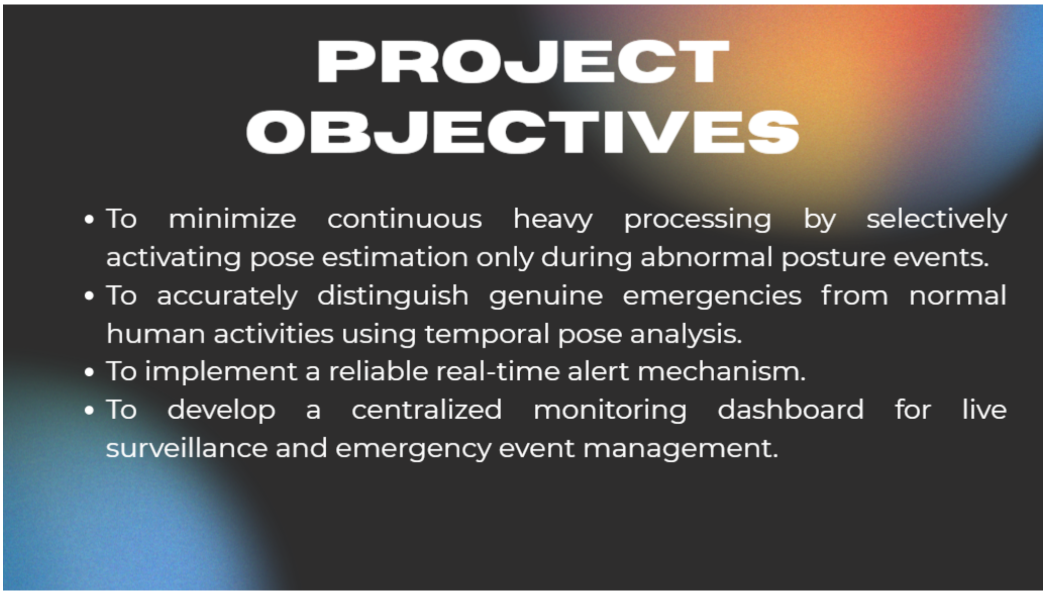
Appendix A: Presentation


QUIETSOS
Guide
Ann Grace Attokaren
Presented by
Gokul Raj
Irene Ajith
Jefry Joseph
Jesel Gibi George


PROBLEM DEFINITION
Continuous pose estimation for emergency detection is computationally heavy and inefficient for real-time surveillance, especially on low-power devices. Therefore, there is a need for a lightweight and intelligent system that selectively performs pose analysis only during abnormal events while reliably detecting genuine emergencies and triggering timely alerts.


PROJECT OBJECTIVES
• To minimize continuous heavy processing by selectively activating pose estimation only during abnormal posture events.
• To accurately distinguish genuine emergencies from normal human activities using temporal pose analysis.
• To implement a reliable real-time alert mechanism.
• To develop a centralized monitoring dashboard for live surveillance and emergency event management.

ASSUMPTIONS

- The system primarily operates in single-person environments, where only one individual is present within the camera's field of view at a time.
- Monitored environments mostly involve upright postures, such as elevator cabins, hospital rooms, library or residential living spaces.
- The camera provides a clear, unobstructed view of the person, with minimal occlusions from furniture or other objects.

ASSUMPTIONS

- Lighting conditions are assumed to be sufficient and stable for reliable human detection and pose estimation.
- The person remains within the camera frame long enough for the system to observe posture changes and temporal motion patterns.
- Normal activities primarily include sitting, standing, bending, or walking, enabling clearer distinction from abnormal events such as falls or faints.

RISKS AND CHALLENGES

Multi-Person Interference:

- The system is optimized for single-person environments; the presence of multiple people may cause confusion in detection and pose tracking.

Occlusion and Visual Obstruction:

- Furniture, walls, or partial camera blockage can hide body parts, reducing pose estimation accuracy during critical events.

Lighting Variations:

- Poor or rapidly changing lighting conditions may degrade human detection and skeletal keypoint extraction performance.

RISKS AND CHALLENGES

Camera Placement Dependency:

- Improper camera angle, height, or distance can negatively affect bounding box detection and posture interpretation.

Edge Device Limitations:

- Although optimized, real-time processing on low-power edge devices may still face performance constraints under high-resolution or multi-camera setups.

Network and Alert Reliability:

- Delays or failures in internet connectivity may impact real-time alert delivery via SMS or voice calls.

LITERATURE SURVEY

Paper	Main Focus	Model Type	Limitations
Human Fall Detection for Smart Home Caring using YOLO Networks	YOLOv5 based static fall detection on custom dataset	YOLOv5n/s/6s object detection	Lacks temporal modeling and pose analysis; processes all frames equally.
A Robust Fall Detection System for Elderly Persons Using YOLOv8	Frame-level fall classification using YOLOv8 with transfer learning	YOLOv8 object detector	Processes all frames; no pose or temporal context.
Fast and High-Precision Human Fall Detection Using Improved YOLOv8 Model	Optimized YOLOv8 with attention modules for high precision and speed	Modified YOLOv8 detector	High accuracy but frame-based only; lacks selective computation.
QuietSOS: Fall Detection and Alerting System	Integrated framework with prefiltering, pose estimation, and temporal sequence modeling using LSTM	YOLOv8 Pose + LSTM hybrid	Selectively processes frames, models temporal dynamics, reduces computation, and improves fall detection accuracy in real time.

DATASET

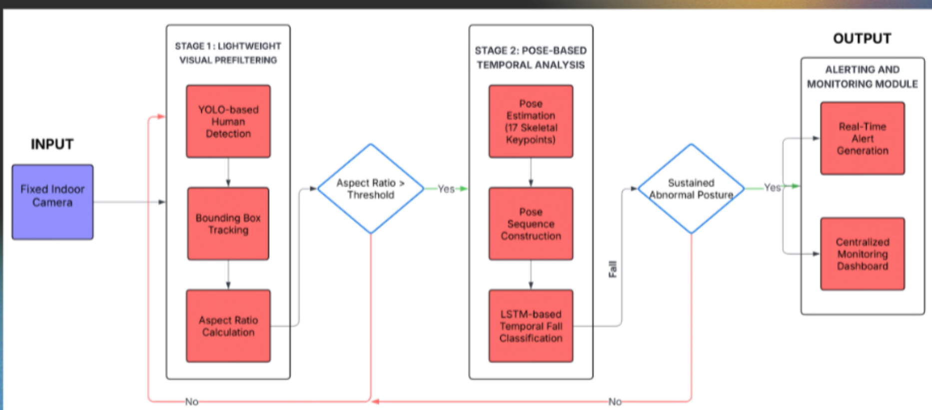
- This dataset contains recorded indoor videos of fall and non-fall activities captured in realistic environments.
- It includes both fall events and normal activities of daily living .
- Pose keypoints stored in labeled CSV files for training and evaluation.

CSV FILE FORMAT

Frame	Keypoint	X	Y	Confidence
1	Nose	887.0295	291.497	0.999612
1	Left Eye	893.7718	276.0556	0.999323
1	Right Eye	893.5267	274.7232	0.999274
1	Left Ear	893.2099	273.1479	0.999496
1	Right Ear	892.4576	275.3683	0.99953
1	Left Shoul	800.7289	296.3414	0.998381
1	Right Shou	801.7031	304.9897	0.999735
1	Left Elbow	812.8003	379.4524	0.092416
1	Right Elbo	822.4731	392.8804	0.981079
1	Left Wrist	859.8165	453.0038	0.156751
1	Right Wris	870.7602	470.7148	0.900513
1	Left Hip	669.9255	425.9225	0.998944
1	Right Hip	660.3382	433.7355	0.999057
1	Left Knee	618.1201	521.5059	0.398187
1	Right Knee	611.1808	533.0822	0.990843
1	Left Ankle	565.6684	631.8463	0.750626
1	Right Ankl	552.0197	651.05	0.993859
2	Nose	903.9511	300.6213	0.999574
2	Left Eye	909.2367	285.5034	0.999271
2	Right Eye	908.5958	284.2949	0.999214
2	Left Ear	907.7038	282.9214	0.999449
2	Right Ear	908.085	285.019	0.999487
2	Left Shoul	811.188	296.3412	0.998429
2	Right Shou	817.6697	308.8322	0.99973
2	Left Elbow	830.2733	379.582	0.088457

- **Frame** – Index of the video frame from which the keypoint data is extracted.
- **Keypoint** – Name of the detected human body joint (e.g., Nose, Shoulder, Knee).
- **X** – Horizontal pixel coordinate of the keypoint in the video frame.
- **Y** – Vertical pixel coordinate of the keypoint in the video frame.
- **Confidence** – Model's confidence score indicating the reliability of the detected keypoint.

ARCHITECTURE DIAGRAM



MODULES

Video Capture Module

- Captures real-time video streams from surveillance cameras deployed in indoor environments such as offices, elevators, hospitals, and residential buildings.

Prefilter Module

- Uses a lightweight YOLO-based prefilter to detect humans and identify abnormal posture or collapse patterns, triggering deeper analysis only when necessary to reduce computational load.

Pose Estimation Module

- Employs YOLOv8Pose to extract accurate skeletal keypoints that represent human posture and movement during suspected emergency situations.

MODULES

Emergency Classification Module

- Utilizes an LSTM network to analyze pose sequences over time and classify genuine emergencies such as falls or faints while filtering out normal activities.

Alerting & Monitoring Dashboard Module

- Sends real-time SMS or voice call alerts via the Twilio API and provides a centralized web-based dashboard for live camera monitoring, instant alert reception, event logging, and secure storage of incident video clips.

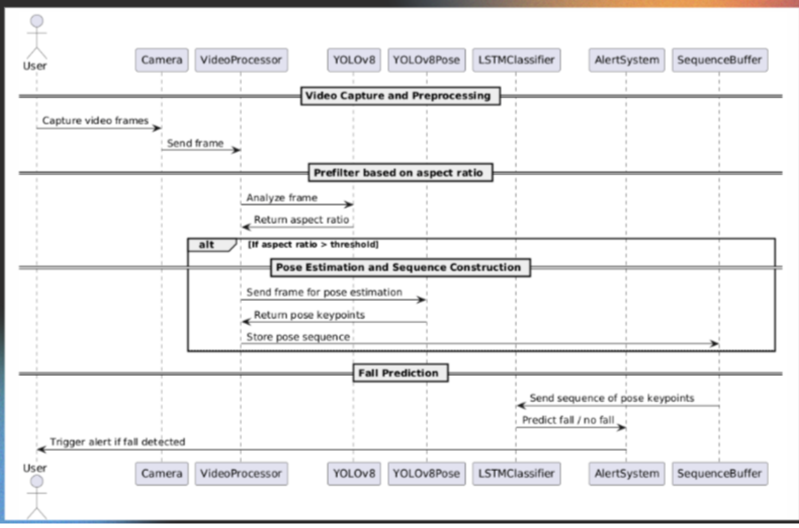
MODEL TRAINING

- Input: Preprocessed pose sequences
  - Shape: 30 time steps × 51 features
- Task: Binary classification (0 - Normal activity, 1 - Fall)
- Dataset split:
  - 80% Training | 10% Validation | 10% Testing
- Class imbalance handled using class weights
- LSTM to capture temporal motion patterns

MODEL TRAINING

- Dropout & Batch Normalization
  - Reduce overfitting & Improve stability
- ReLU highlighting strong fall-related patterns
- Sigmoid converts the model output into a probability indicating the likelihood of a fall.
- Training details:
  - Optimizer: Adam
  - Metrics: Accuracy, Precision, Recall
  - Early stopping ensures best-performing model

SEQUENCE DIAGRAM



PREPROCESSING IN REAL TIME MONITORING

<u>Image Resizing</u>

- Frames are resized internally by YOLOv8 during inference.
- Input resolution used: 320 × 320 pixels

<u>Aspect Ratio Calculation</u>

- Width-to-height ratio (w/h) of the detected person bounding box used as a posture indicator for identifying abnormal body orientation.

<u>Aspect Ratio Normalization (Sigmoid Scoring)</u>

- The raw ratio is converted into a normalized score using a sigmoid function.
- Converts raw ratios into stable probability-like scores.

PREPROCESSING IN REAL TIME MONITORING

EMA Smoothing of Aspect Ratio

- Exponential Moving Average (EMA) smoothing is applied to aspect ratio scores.
- Reduces noise from unstable bounding boxes.
- Prevents false triggers caused by a single noisy frame.

Pose Feature Vector Creation

- When pose estimation is triggered, YOLOv8 Pose extracts body keypoints.
- Each keypoint provides: X, Y,Confidence
- For 17 keypoints → 51 features per frame.

PREPROCESSING IN LSTM TRAINING PIPELINE

Feature Vector Construction

- Each keypoint contributes X, Y, Confidence

Total features per frame: 17 keypoints × 3 features = 51 features

Handling Missing Keypoints

- If a keypoint is missing in a frame, values are replaced with 0, 0, 0
- Maintains consistent feature vector size across all frames.

Sequence Generation

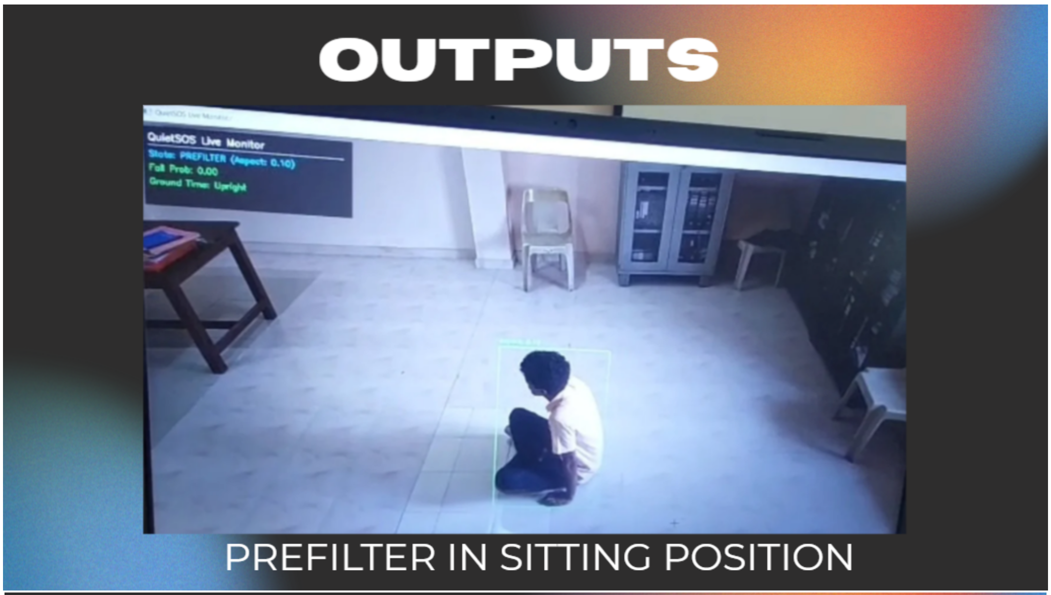
- Frame feature vectors are grouped into temporal sequences:
- Sequence length = 30 frames, Stride = 5 frames

Input Shape Formatting

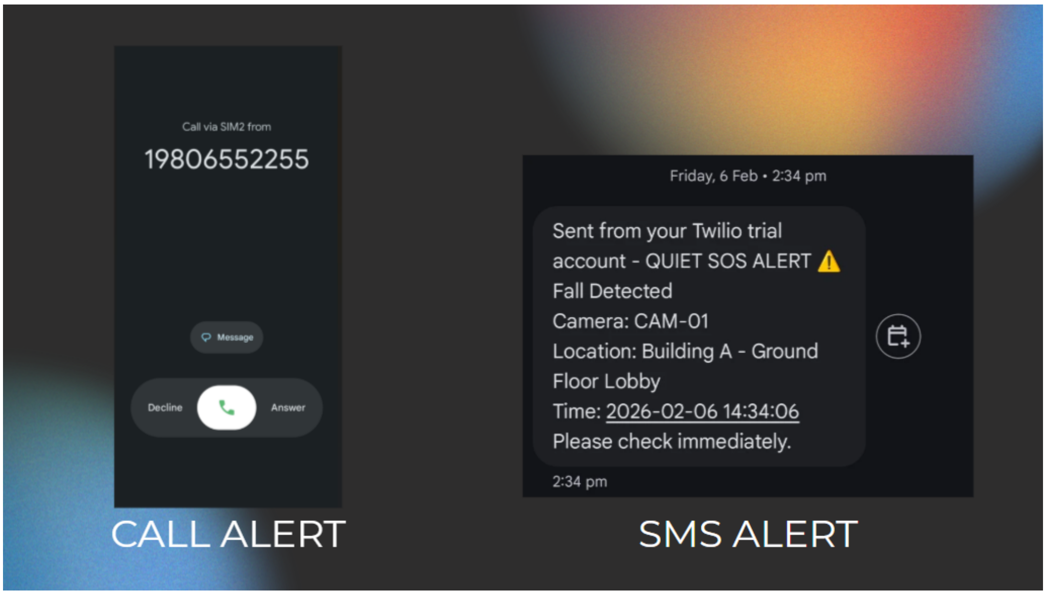
- Final dataset format for the Long Short-Term Memory model:
- (samples, 30, 51)
- 30 → time steps , 51 → pose features

OUTPUTS

PREFILTER IN STANDING POSITION


OUTPUTS
QuietSOS Live Monitor
State: PREFILTER (Aspect: 0.10)
Fall Prob: 0.00
Ground Time: Upright
PREFILTER IN SITTING POSITION


OUTPUTS
QuietSOS Live Monitor
State: LSTM POSE ACTIVE
Fall Prob: 0.76
Ground Time: 10.0s
FALL DETECTED: 10.0s
FALL DETECTED


Call via SIM2 from
19806552255
Message
Decline
Answer
Friday, 6 Feb • 2:34 pm
Sent from your Twilio trial account - QUIET SOS ALERT ⚠
Fall Detected
Camera: CAM-01
Location: Building A - Ground Floor Lobby
Time: 2026-02-06 14:34:06
Please check immediately.
2:34 pm
CALL ALERT
SMS ALERT


Fall Detection System
System Active
Last 24 Hours
Back to QuietSOS
Active Cameras
12
All operational
+2 from yesterday
Total Alerts Today
3
2 resolved
-1 from yesterday
Real-time Alerts
1 Active Alert
• Critical
✓ Resolved
✓ Resolved
0:00
DASHBOARD

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

1. Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop solutions to complex engineering problems.

2. Problem analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

3. Design/development of solutions: Design creative solutions for complex engineering problems and design or develop systems, components or processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment.

4. Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.

5. Engineering tool usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, recognizing their limitations to solve complex engineering problems.

6. The engineer and the world: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

7. Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.

8. Individual and collaborative teamwork: Function effectively as an individual and as a member or leader in diverse or multidisciplinary teams.

9. Communication: Communicate effectively and inclusively within the engineering community and society at large, including the ability to comprehend and write effective reports and design documentation, and make effective presentations considering cultural, language and learning differences.

10. Project management and finance: Apply knowledge and understanding of engineering management principles and economic decision-making to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

11. Life-long learning: Recognize the need for and have the preparation and ability for independent and life-long learning, adaptability to new and emerging technologies, and critical thinking in the context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills
The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills
The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills
The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

After the completion of the course the student will be able to:

Course Outcome 1: Model and solve real-world problems by applying knowledge across domains (Cognitive knowledge level: Apply).

Course Outcome 2: Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

Course Outcome 3: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).

Course Outcome 4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

Course Outcome 5: Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: Analyze).

Course Outcome 6: Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES:

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND COURSE OUTCOMES (COs) – PROGRAM SPECIFIC OUTCOMES (PSOs)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO-PO MAPPING

MAPPING	LOW/ MEDIUM / HIGH	JUSTIFICATION
101003/ CS822U.CO1-PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1-PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1-PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1-PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1-PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1-PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1-PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.

101003/ CS822U.CO1-PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1-PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1-PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1-PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/ CS822U.CO1-PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1-PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2-PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/ CS822U.CO2-PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2-PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.

101003/ CS822U.CO2-PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2-PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2-PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2-PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2-PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2-PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2-PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2-PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2-PSO2	Medium	Uses programming and software development practices to implement solutions.

101003/ CS822U.CO3-PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3-PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3-PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3-PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3-PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/ CS822U.CO4-PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4-PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4-PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4-PO6	Medium	Considers societal and environmental impact while implementing solutions.

101003/ CS822U.CO4-PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4-PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4-PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4-PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4-PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4-PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4-PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5-PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5-PO2	High	Requires deep problem analysis and identification of research gaps.

101003/ CS822U.CO5-PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5-PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5-PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5-PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5-PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5-PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6-PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6-PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6-PO7	Medium	Requires responsible and ethical communication of technical results.

101003/ CS822U.CO6-PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6-PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6-PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6-PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6-PSO3	High	Communication of innovative computer science solutions and research outcomes.